Gimbal Mount Assembly
Interface Control Document

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1. Introduction

1.1. Scope

This document identifies and defines the internal and external interfaces of the Gimbal Mount Assembly (GMA).

1.2. Reference

[RD1] Nyx Moon - Huracan Development Logic / TEC-FRA-DOC-01004 / Issue 02

[RD2] PCB PIEZOTRONICS / General Operating Guide / 2002

[RD3] Gimbal Mount Assembly - Definition File / Version 001

2. Overview

2.1. Engine System and GMA

The figure shows the location of the GMA within the engine system, progressing from the complete engine assembly to a detailed view of the GMA.

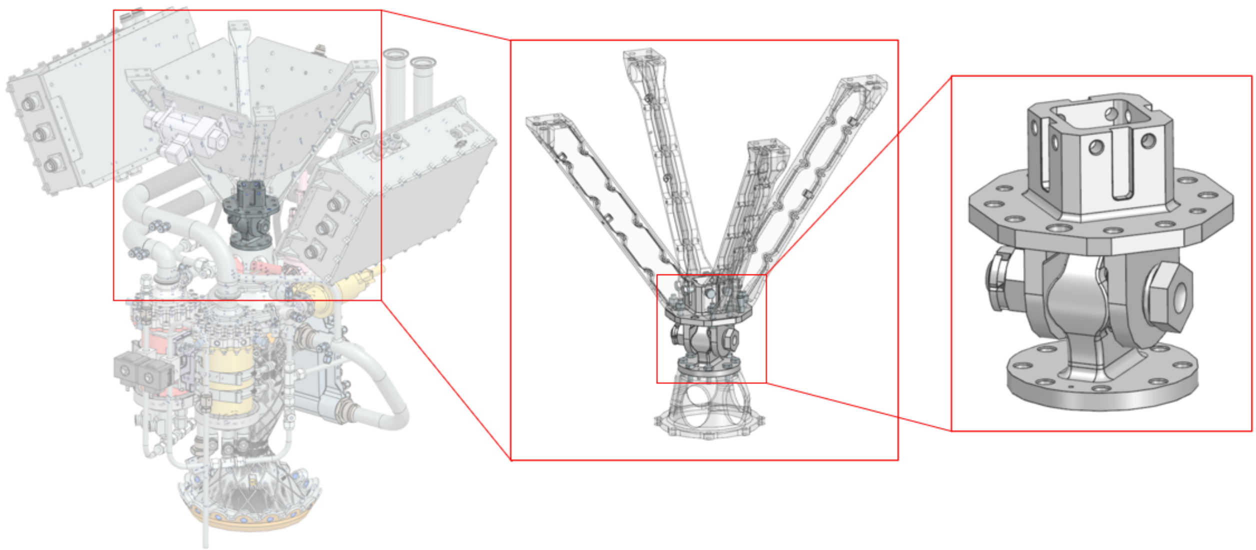

Figure 1: GMA location within the engine system

3. Interfaces

3.1. Internal Interfaces: GMA

The internal interfaces comprise all joints that connect the GMA component. As shown in the exploded view below, the primary internal mechanical interface is the connection between the GMA Lug and Clevis Head. The joint is established by a single bolt-and-nut assembly and secured against loosening by a Cotter Pin.

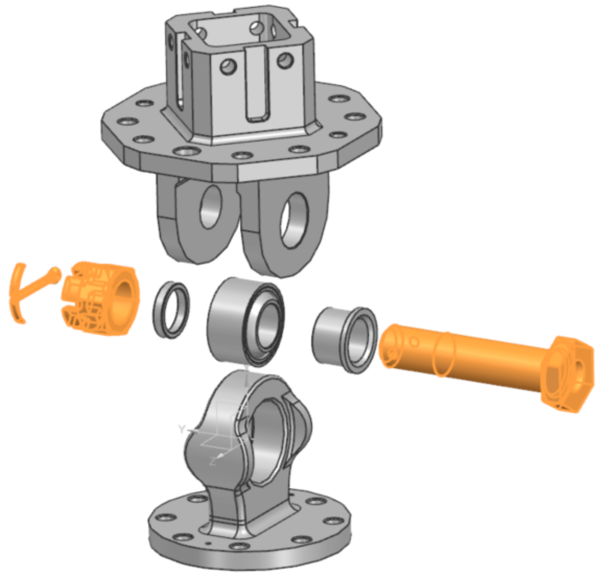

Figure 2: GMA Internal Interface

The controlled interface components and their material specifications are listed in Table 1.

Table 1: GMA Internal Interfaces

Interface Part	Material	Quantity
Bolt NAS6710DU29	A286	1
Nut MS9358-16	A286	1
Cotter Pin MS24665-374	AISI 302 or 304	1

3.2. External Interfaces: GMA and Engine/Vehicle

The external interfaces encompass all joints connecting the GMA to the engine on one side and to the vehicle on the other.

Engine Interface
The GMA is attached to the Thrust Dome of the Thrust Chamber Assembly (TCA), as defined in [RD3]. Eight stainless-steel fasteners shall secure the interface and shall be distributed symmetrically around the flange. Two stainless-steel dowel pins shall provide interface alignment.

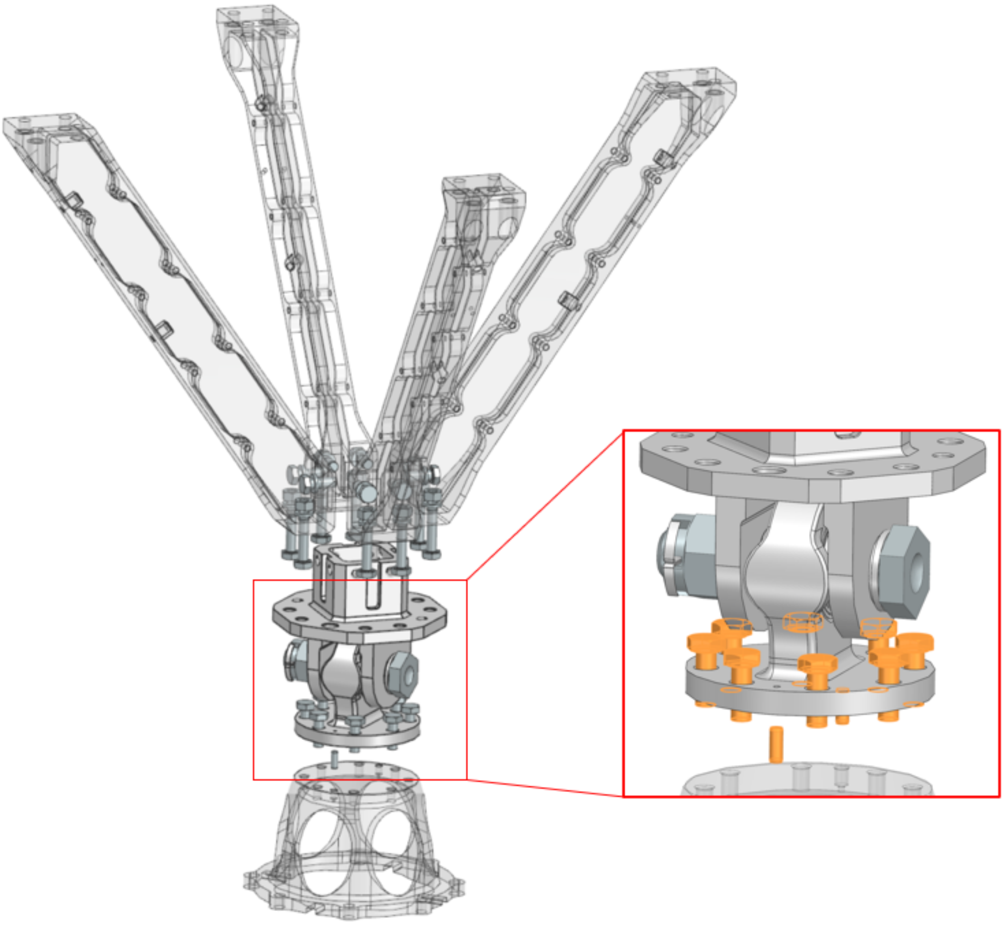

Figure 3: GMA-to-engine interface at the Thrust Domee

Table 2: GMA-to-engine interface hardware

Interface Part	Designation	Quantity
Hexagon head screw	ISO4017-M6x16-A4-70	8
Dowel pin	ISO8734-4x10-C1	2

Vehicle Interface

The GMA interfaces with the vehicle structure through four Thrust Frame Beams, as defined in [RD3]. Each Thrust Frame Beam interfaces with two M6 threaded holes, two Ø6.6 mm through-holes, and one centering feature on the GMA Clevis Head. Across all four Beams, the interface therefore comprises eight M6 threaded holes, eight Ø6.6 mm through-holes, and four centering features. The centering features align the Beams with the Clevis Head before the fasteners are tightened.

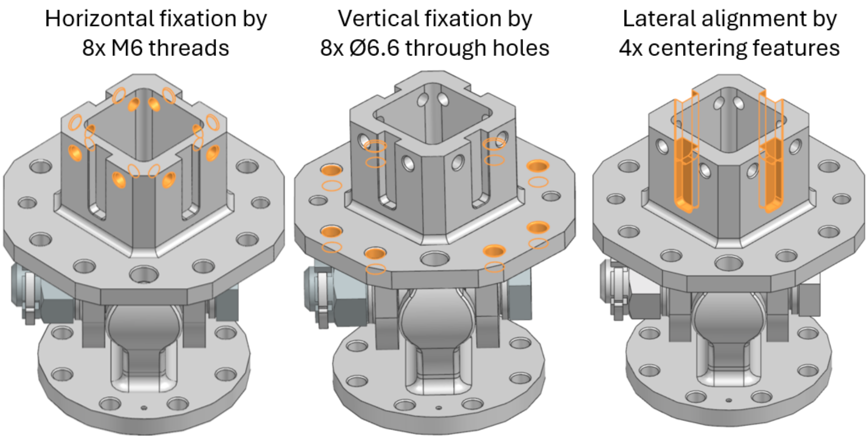

Figure 4: Interfaces between GMA and Thrust Frame Beams

Table 3 show the total number of bolts, nuts, and washers required for the four thrust frame beams.

Table 3: GMA-to-vehicle interface hardware

Interface Part	Designation	Quantity
Hexagon head screw	ISO4017-M6x16-A4-70	8
Hexagon head screw	ISO4017-M6x25-A4-70	8
Hexagon regular nut	ISO4032-M6-A4-70	8
Plain washer	ISO7092-6-200HV-A4	16

Plain washers shall be used wherever the fasteners bear against softer materials, such as the aluminum Thrust Frame Beams. An overall view of the GMA with its adjacent parts to the engine respectively vehicle and fasteners is shown below.

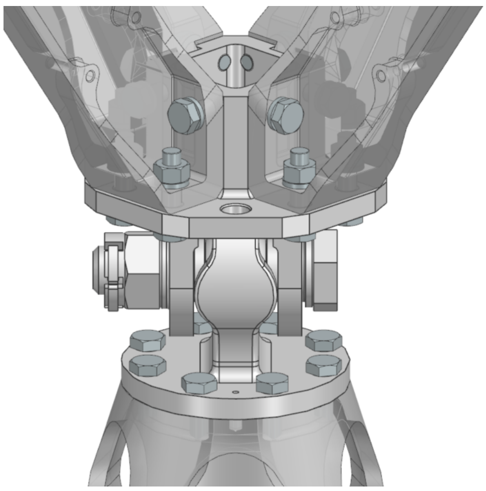

Figure 5: Overall GMA installation with engine- and vehicle-side interface hardware

3.3. Handling and Transport Interfaces

An M8 stainless-steel eyebolt shall be used for handling operations. Two symmetrically arranged Ø9 mm through-holes are provided in the GMA clevis-head flange, as shown in Figure 6.

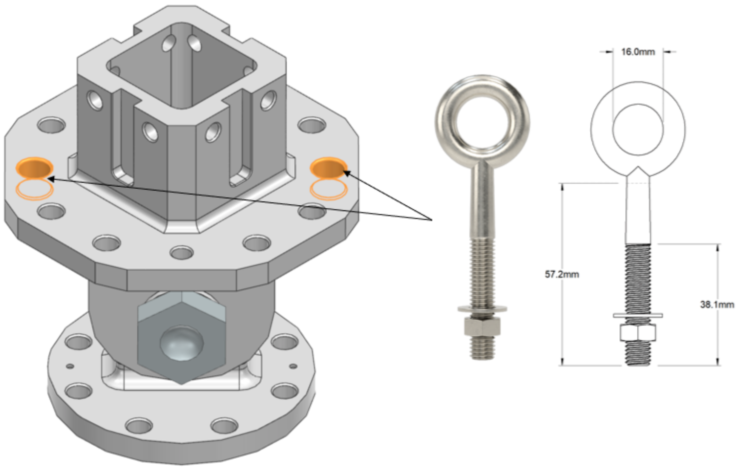

Figure 6: Interface for handling operations

Table 4: Interface for GMA handling

Interface type	Size	Material	Quantity
Eyebolt	M8	AISI 316	2

3.4. Instrumentation Interface

For the current design state, a shock sensor is shall to be attached on the GMA. Three alternative 1/4-28 UNF-3B threaded mounting locations are provided on the GMA Clevis-Head flange for installation of the shock accelerometer shown in Figure 7.

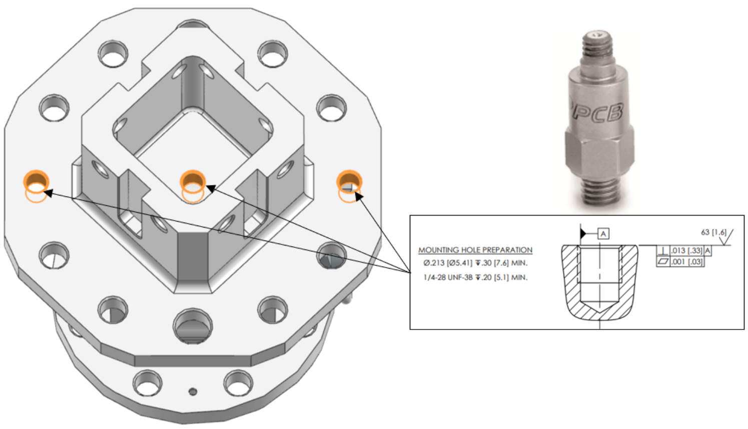

Figure 7: Shock-accelerometer mounting interface

Table 5: Interface for GMA instrumentation

Interface type	Manufacturer	Model	GMA Interface
Shock Accelerometer	PCB PIEZOTRONICS	305C3	1/4-28 UNF-3B

4. Annex

4.1. Manufacturing drawings

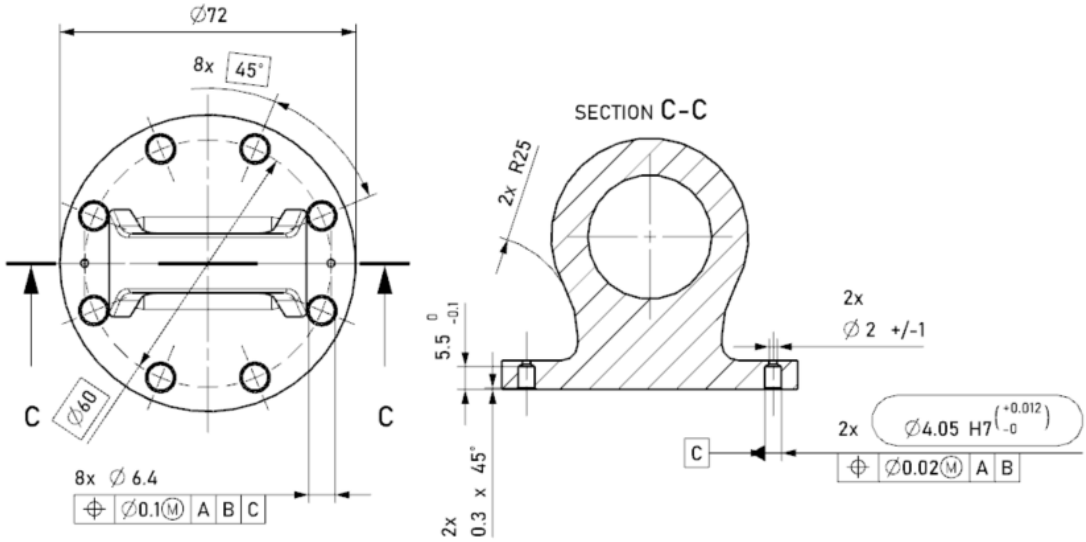

Figure 8: Drawing of GMA Lug Head interface

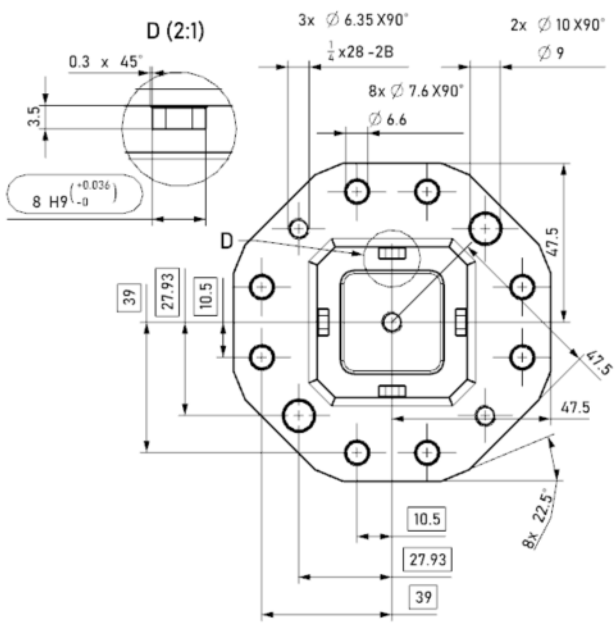

Figure 9: Drawing of GMA Clevis Head interface

5. Acronym List

The acronyms used in this document are listed below.

Table 6: Acronyms

Acronym	**Definition**
GMA	Gimbal Mount Assembly
TCA	Thrust Chamber Assembly